

L5 Planning Quiz

1. What is the primary output of mission planning in an autonomous-driving system?

- A) A map-level route from the start to the destination
- B) A steering-angle command
- C) A set of detected objects
- D) A throttle and brake signal

ANS:

2. What is the primary output of behavior planning?

- A) A raw occupancy grid
- B) A selected high-level driving maneuver, such as stop, follow, or change lanes
- C) A sequence of actuator voltages
- D) A global HD map

ANS:

3. What does motion planning generate for the vehicle controller to track?

- A) A traffic-rule database
- B) A destination address
- C) A feasible reference trajectory
- D) A sensor-calibration table

ANS:

4. In route planning, the road network is commonly represented as which structure?

- A) An unweighted image
- B) A finite-state controller
- C) A speed profile
- D) A weighted graph

ANS:

5. What problem does route planning solve?

- A) Finding an optimal navigation path from start to goal
- B) Predicting the color of a traffic light
- C) Estimating tire pressure
- D) Selecting a camera exposure

ANS:

6. Which value does Dijkstra's algorithm use to prioritize a node?

- A) Only the estimated cost from that node to the goal
- B) The currently known cost from the start to that node
- C) The node's geographic latitude
- D) The number of outgoing edges only

ANS:

7. In A* search, what does the evaluation value normally combine?

- A) Vehicle speed and steering angle

- B) Road width and lane count
- C) Cost from the start and estimated cost to the goal
- D) Camera and LiDAR confidence

ANS:

8. What property must a heuristic have to be admissible for A* optimality?

- A) It must always equal zero
- B) It must overestimate the remaining cost
- C) It must ignore the goal
- D) It must never overestimate the true remaining cost

ANS:

9. Why can A* explore fewer positions than Dijkstra's algorithm?

- A) Its heuristic directs the search toward the target
- B) It ignores all edge costs
- C) It expands nodes randomly
- D) It searches only backward from the goal

ANS:

10. Which information should a behavior planner consider when choosing a driving maneuver?

- A) Only the destination coordinates
- B) Road rules plus static and dynamic objects
- C) Only the current steering angle
- D) Only the vehicle's battery level

ANS:

11. What does the 'Follow Leader' maneuver require the ego vehicle to do?

- A) Pass the leading vehicle immediately
- B) Stop regardless of traffic
- C) Match the leader's speed while maintaining a safe distance
- D) Ignore the leading vehicle

ANS:

12. In a finite-state-machine behavior planner, what normally triggers a transition between states?

- A) A randomly selected timer
- B) A software recompilation
- C) A change in map file format
- D) An input condition or event

ANS:

13. How does reinforcement-learning-based behavior planning learn a policy?

- A) Through trial and error to maximize cumulative reward
- B) By storing only hand-written traffic rules
- C) By applying Dijkstra's algorithm to every maneuver

D) By eliminating all environmental feedback

ANS:

14. Why might an FSM use $Ego.Velocity \leq StopThreshold$ instead of $Ego.Velocity == 0$?

A) To increase the maximum speed

B) To improve robustness to state-estimation noise

C) To disable stopping behavior

D) To remove the need for sensors

ANS:

15. Why is there no direct transition from 'Decelerate to Stop' to 'Track Speed' at a stop-sign intersection?

A) The vehicle must always turn left

B) The intersection has no approaching zone

C) The vehicle must first come to a complete stop

D) The behavior planner cannot measure speed

ANS:

16. How does a hierarchical FSM organize multiple driving scenarios?

A) It uses one state for all situations

B) It removes all entry and exit transitions

C) It replaces behavior planning with route planning

D) It models each scenario as a super-state containing a scenario-specific lower-level FSM

ANS:

17. Which is a major limitation of FSM-based behavior planning?

A) State explosion as scenarios become numerous and complex

B) Inability to represent any driving maneuver

C) Requirement that every road be straight

D) Inability to use logical conditions

ANS:

18. How is a trajectory distinguished from a geometric path?

A) A trajectory contains no spatial information

B) A trajectory is a path parameterized by time

C) A trajectory is always a straight line

D) A trajectory contains only map-level turns

ANS:

19. Into which two main tasks can motion planning be decomposed?

A) Localization and mapping

B) Detection and segmentation

C) Path planning and speed-profile generation

D) Mission planning and sensor fusion

ANS:

20. What does an occupancy grid encode?

- A) Only the vehicle's control gains
- B) Only the destination and start points
- C) Only dynamic-object identities
- D) Occupied and free regions in a discretized environment

ANS:

21. What is a key difference between RRT and PRM when start and goal locations change?

- A) RRT typically grows a new graph, whereas PRM can reuse its roadmap
- B) PRM cannot represent obstacles
- C) RRT always reuses one fixed roadmap
- D) PRM does not require path search

ANS:

22. After constructing a probabilistic roadmap, which algorithms can be used to find a route through it?

- A) PID or MPC
- B) A* or Dijkstra's algorithm
- C) K-means or PCA
- D) Backpropagation or gradient descent

ANS:

23. What does it mean that many RRT and PRM variants are probabilistically complete?

- A) They always return the shortest path
- B) Their runtime is fixed and predictable
- C) If a path exists, it will be found in finite time, although the time may be very long
- D) They never require collision checking

ANS:

24. Why is path smoothing often applied after sampling-based planning?

- A) To add more collisions to the route
- B) To convert the route into a global map
- C) To increase unnecessary turns
- D) To reduce path length and produce a smoother, easier-to-track path

ANS:

25. What is the function of speed-profile generation?

- A) Assigning a speed value to every point along a path
- B) Selecting the destination city
- C) Classifying road signs
- D) Building the occupancy grid

ANS:

26. What characterizes a linear-ramp speed profile?

- A) Constant speed at all times

- B) Constant acceleration until the reference speed is reached
- C) Random acceleration and braking
- D) A mandatory full stop at every path point

ANS:

27. Which profile is described as constant-zero-constant deceleration and can be used to stop at a red light?

- A) A circular profile
- B) A linear-ramp profile
- C) A trapezoidal profile
- D) A probabilistic roadmap

ANS:

28. Why is 'avoid collisions at all costs' not a practical autonomous-driving objective?

- A) It requires all vehicles to drive faster
- B) It eliminates the need for safety rules
- C) It guarantees the AV is responsible for every crash
- D) It can make the system so conservative that it becomes useless

ANS:

29. What is the central safety goal of Responsibility-Sensitive Safety (RSS)?

- A) The AV should not cause accidents and should properly respond to others' mistakes
- B) The AV should never be involved in any collision under any circumstances
- C) The AV should always claim right-of-way
- D) The AV should ignore implicit traffic rules

ANS:

30. In RSS, what is 'blame time'?

- A) The moment an insurance claim is filed
- B) The first moment when a situation becomes dangerous
- C) The time at which route planning begins
- D) The delay between two sensor frames

ANS: